

A Low-Cost Microclimate Monitoring System for Orchard Disease Management

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Orchard Disease Management



Southern Blight
Source: K. Peter



Powdery Mildew
Photo by K. Peter



Apple Scab
Photo by K. Peter



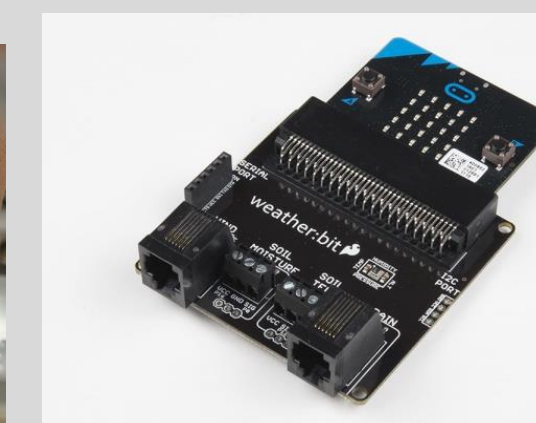
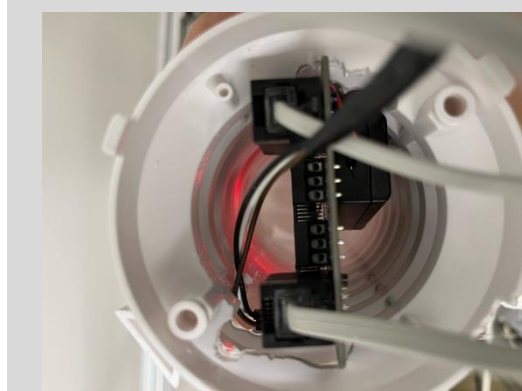
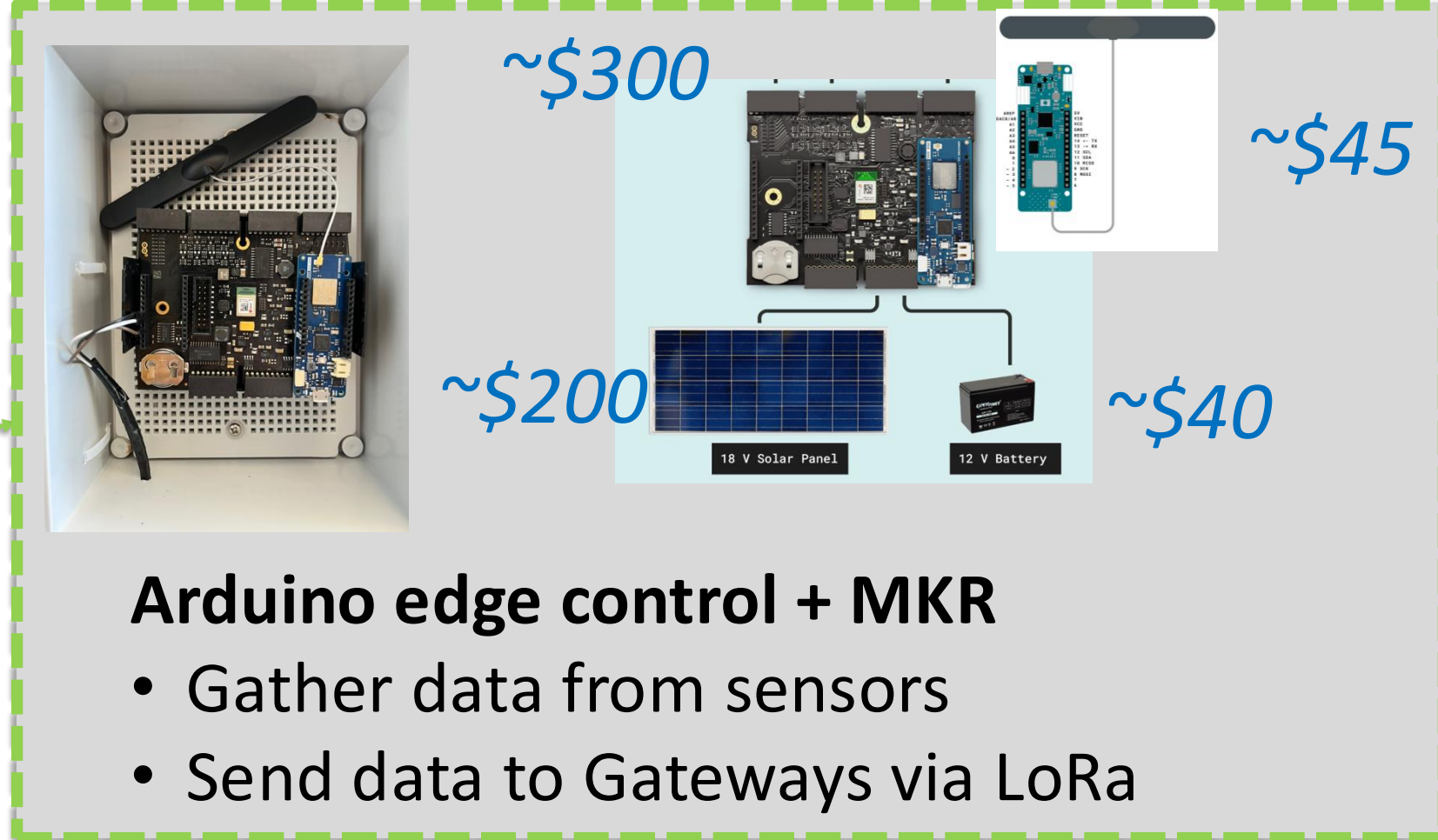
Weather station
METER

- Disease severity and type depend on environment conditions.
- Orchard microclimate varies from local weather.
- Orchard real-time climate monitoring helps disease management.
- Wireless All-In-One Weather Stations and **services** are expensive.

Low-cost weather station



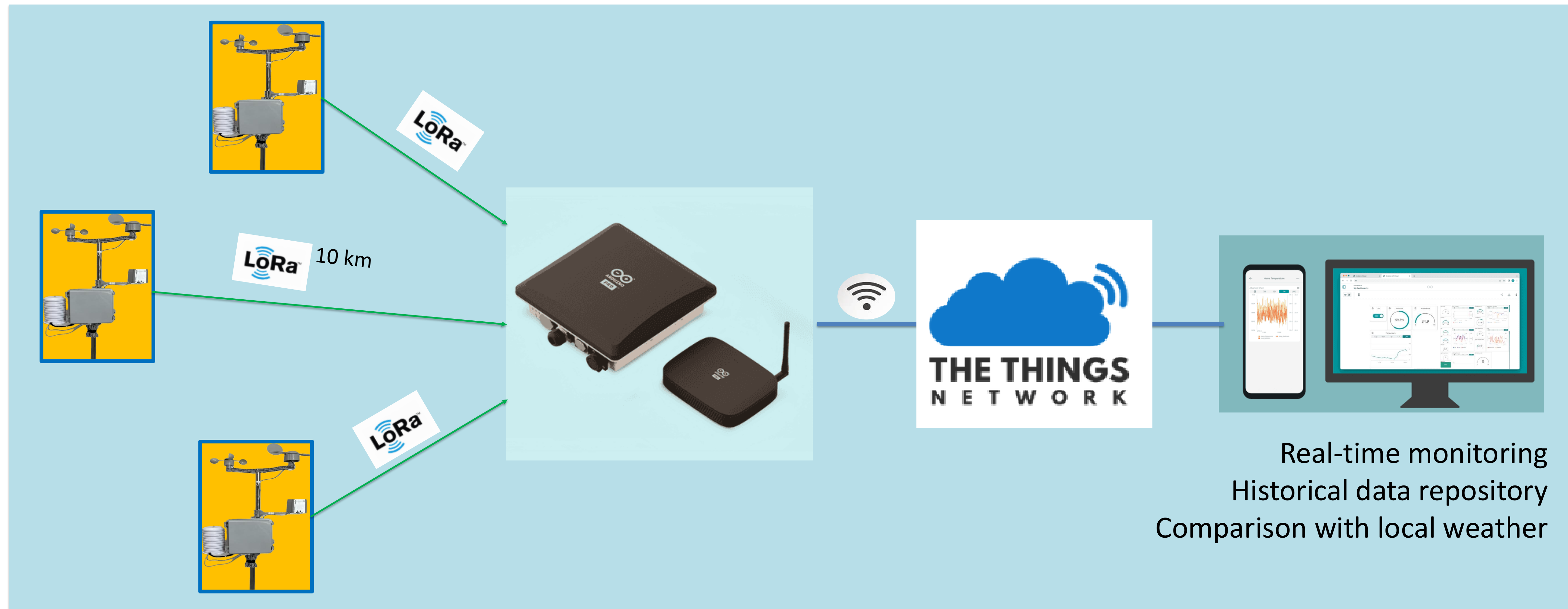
- Wind speed
- Wind direction
- Rainfall
- ~\$80



~\$50

- Measure barometric pressure, relative humidity and temperature
- Gather wind and rain readings
- Wired connection with Arduino edge control
- Optional connectors for soil moisture and soil temperature

LoRaWAN network





~\$600

Gateway

- Receive data sent by MKR
- Send data to the cloud
- Placed at office/house
- Serve for many sensor nodes

Low-cost!

Cost

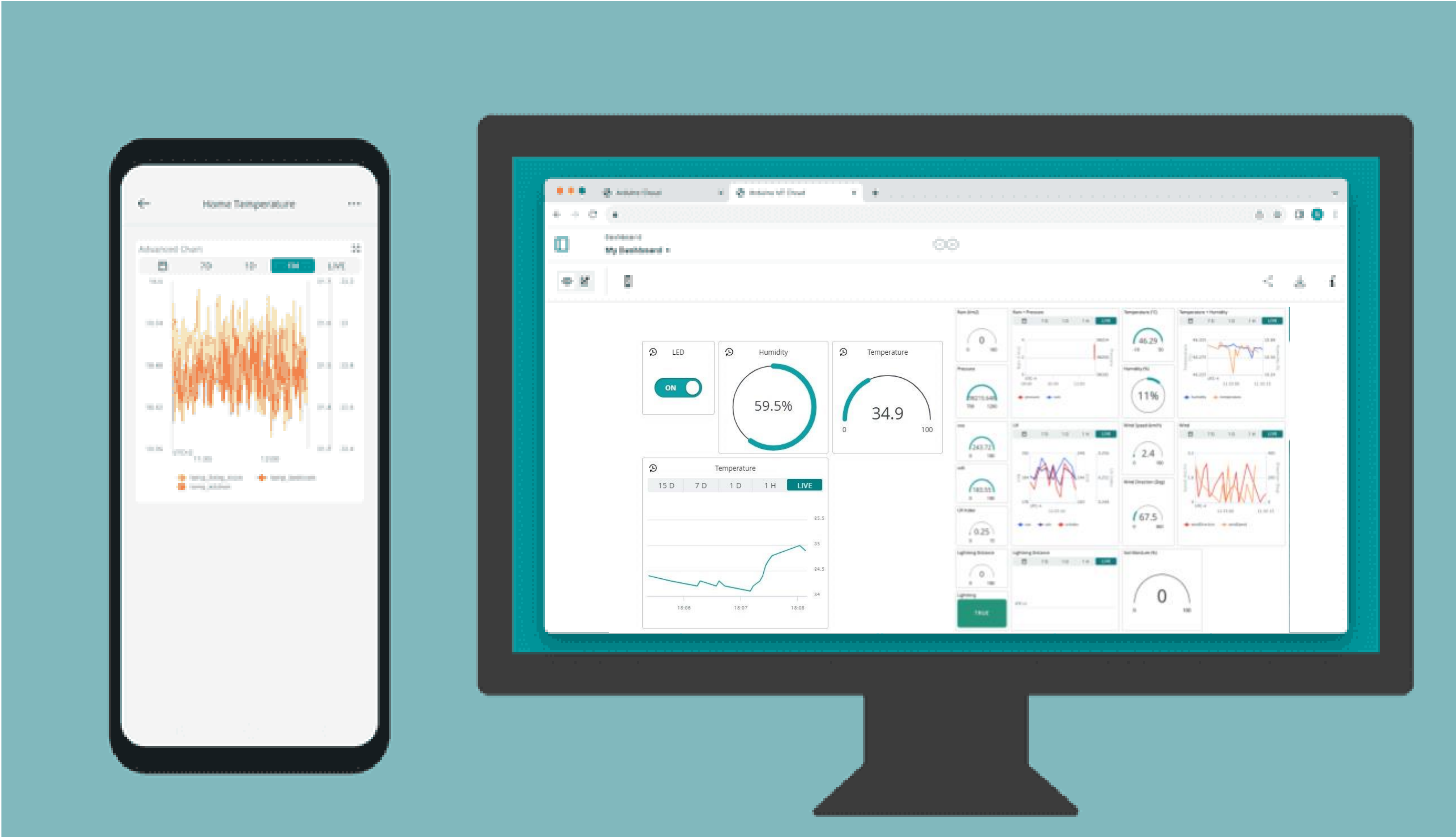
- One weather station: \$715
- One gateway: \$600
- 1 gateway + 3 stations: \$2745
- No subscription cost

What's on the market



METER Group Weather station *\$1,850*

A web platform



Field test



Acknowledgement



STATE HORTICULTURAL ASSOCIATION OF
PENNSYLVANIA